•Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU

2026

Content

- Basic Concepts
- Dispersion Compensation Management

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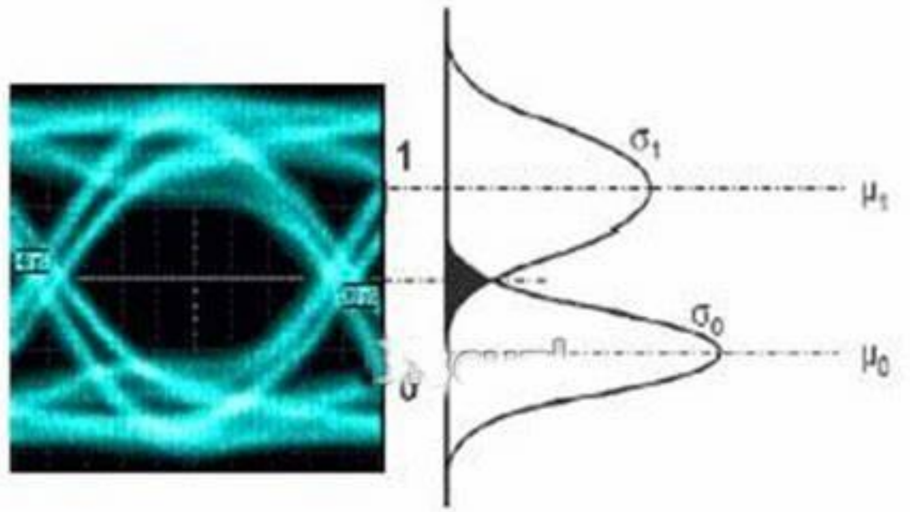
- Dispersion Compensation Management

Basic Optical Network

Digital Communications： 0/1 Signals

Questions：

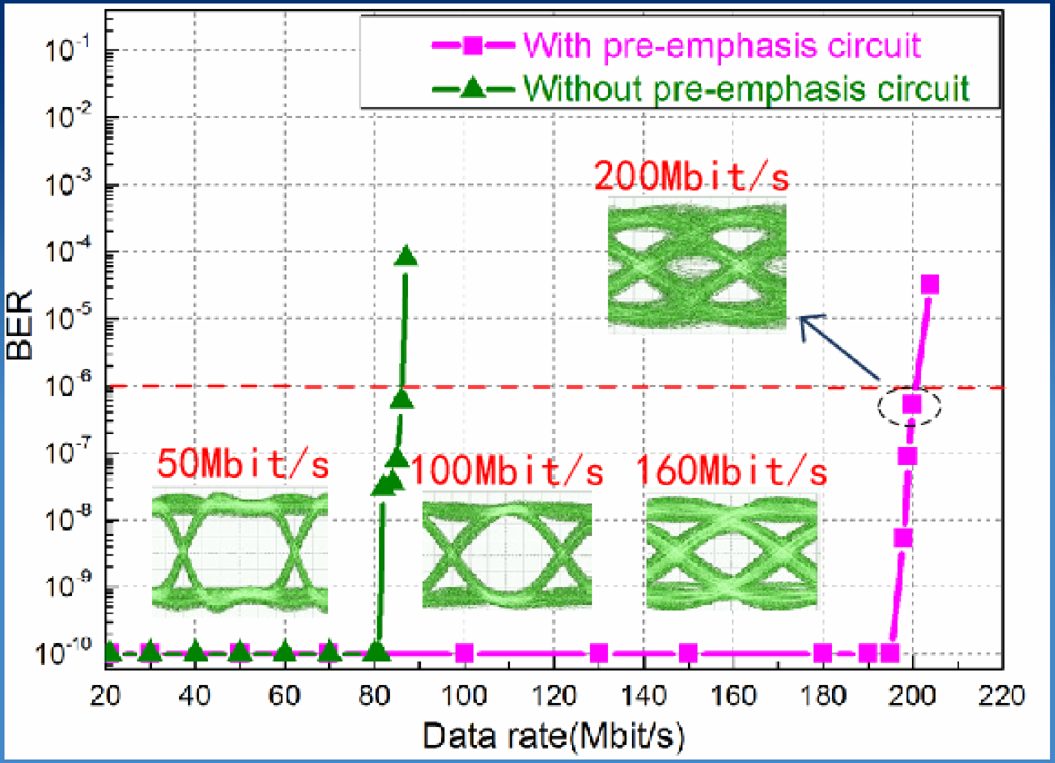
- Practically, what is 1 or 0?
- 1 or 0 is not a value, but a distribution.
- Overlap between regions of 1 and 0 is unavoidable, question is how much
- BER： bit error rate quantifies this overlap region.
- Direct measurement of BER is very time-consuming, using Q factor instead.



$$Q-factor = \frac{\mu_1 - \mu_0}{\sigma_1 + \sigma_0}$$

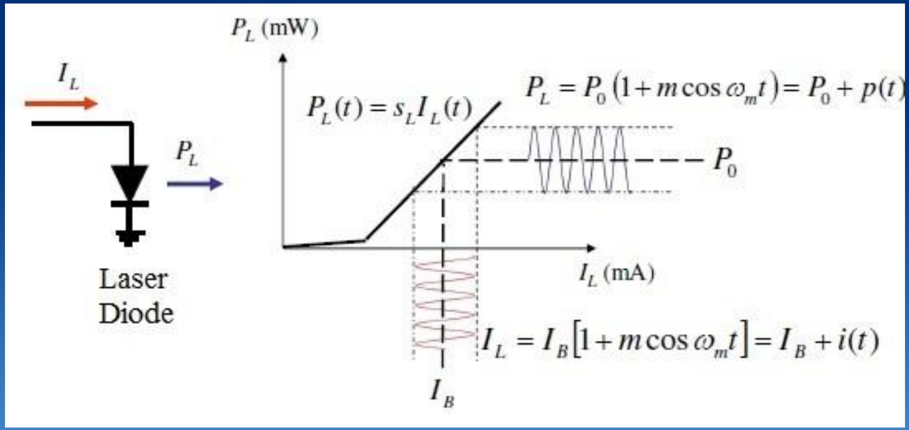
BER v.s. data rates

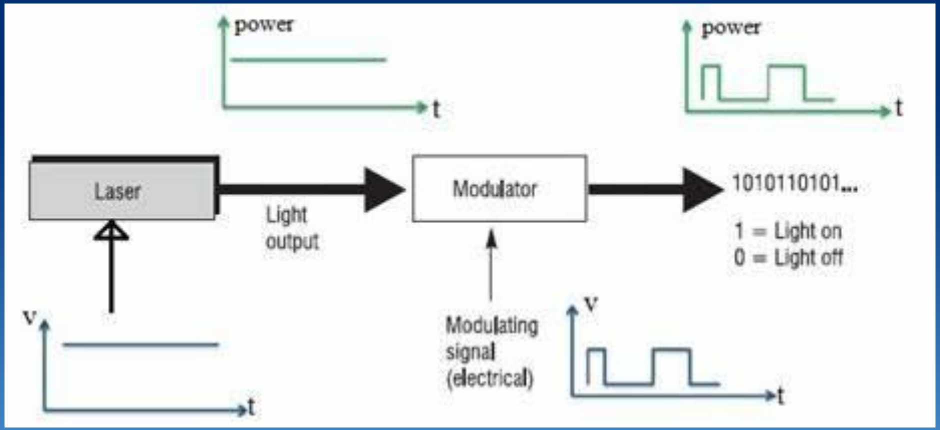
- Eye diagrams deteriorate with data rates, so are BERs.



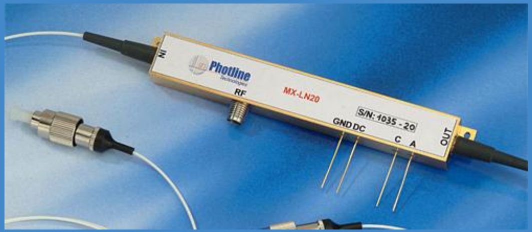
Generation of 1/0 signal: modulation of light

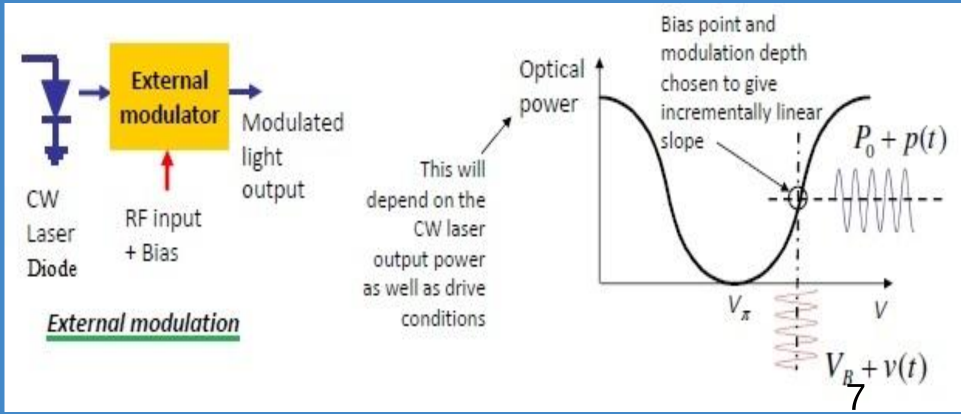
- Modulation: direct or external modulation





- Simple, slow-rate, low-cost
- Additional device needed, fast-rate





•A look at BER again

- **Definition:**

The number of Bit errors that occur within the space of one second.

- **Range: 10^{-9} to 10^{-12}**

The higher the Data Transmission rate the greater the standard.

✓ **DS-1** (1.544 Mbit/s) **signal is considered acceptable with a BER of 10^{-6},**

✓ **OC-3 (**155.52 Mbit/s) **signal requires a BER of no more than 10^{-12}.**

•A look at BER again

- **Test equipment:** **BERT (BER tester)**

  two fundamental parts:

  1. **signal pattern generator**
  2. **error detector**

- **Signal pattern generator**
  - **Producing a known data sequence, intentionally stress some aspect, like difficulty for the clock recovery system to synchronize.**
  - **Most common pattern: pseudo-random binary sequence (PRBS)**

- **Error detector**
  - **compared bit by bit between two signals.**
  - **BER = number of incorrectly received bits / total number of bits transmitted**

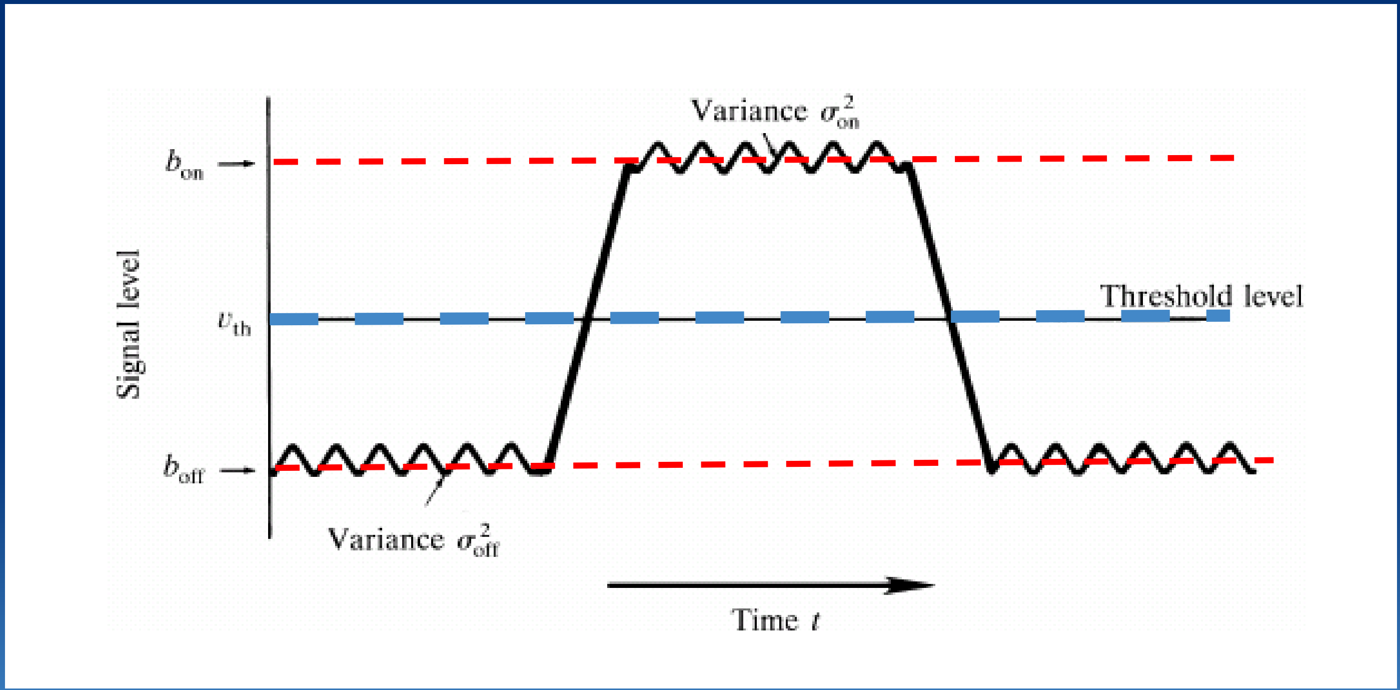
•Q factor and BER

- **Why is BER difficult to simulate or calculate?**
  - **For a given design BER $<10^{-12}$ (OC-3, or 155 Mbps), the network would have one error in approximately 10 days. It would take 1000 days!! to record a steady state BER value!**
  - **Example: Assuming a confidence level of 99%, BER threshold set at 10^{-10} and a bit rate of 2.5 Gb/s, the required sample number n is 6.64×10^{10}**
- **Q-factor analysis is comparatively easy.**
  - **Q -> estimate BER**
  - **dynamically calculate Q from OSNR, eye-diagram**

$$Q = \frac{I_1 - I_0}{\sigma_1 + \sigma_0}$$

$$BER = \frac{1}{2} erfc\left(\frac{Q}{\sqrt{2}}\right)$$

In the format of threshold and on-off levels



When the probabilities of 0 and 1 pulses are equal, the bit error rate is given by

$$BER = \frac{N_{error}}{N_{total}} = P_e(Q) = \frac{1}{\sqrt{\pi}} \int_{Q/\sqrt{2}}^{\infty} e^{-x^2} dx = \frac{1}{2}\left[1 - erf(\frac{Q}{\sqrt{2}})\right]$$

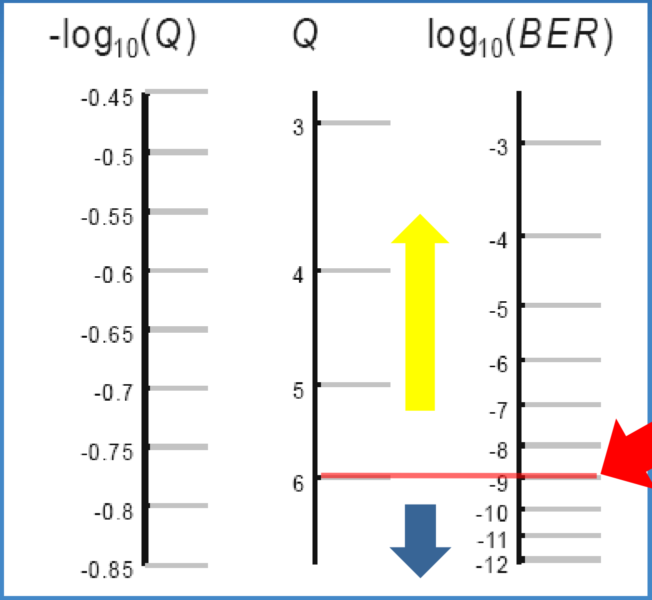
$$\approx \frac{2}{\sqrt{2\pi}} \frac{e^{-Q^2/2}}{Q} \qquad (7)$$

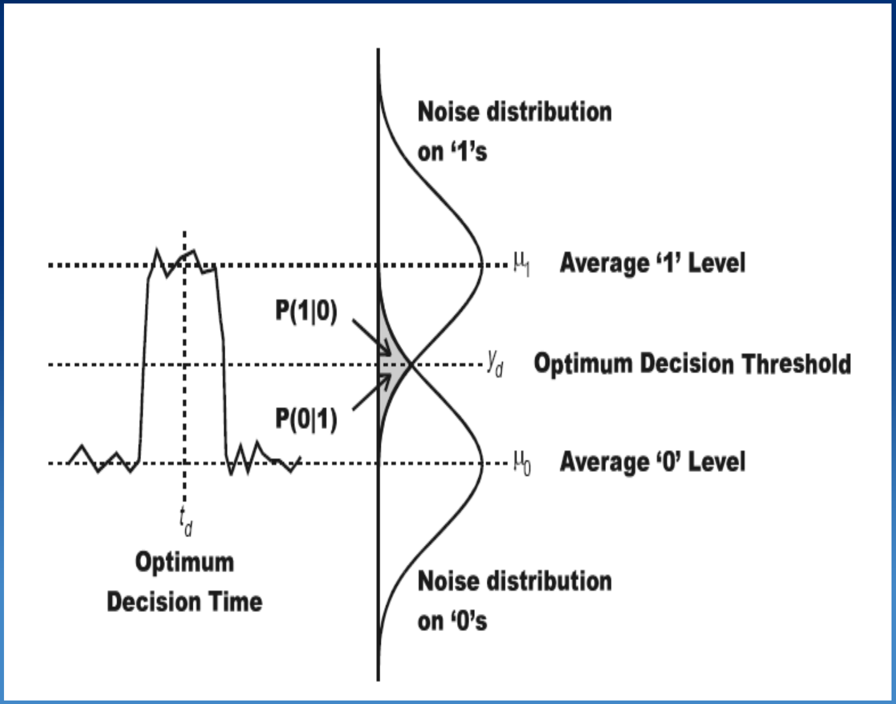
where

$$Q = \frac{V_{th} - b_{off}}{\sigma_{off}} = \frac{b_{ofn} - V_{th}}{\sigma_{on}}$$

•Q factor and BER

- **Q is a measure of the quality of any signal**
  - **If the noise is known to be Gaussian, Q fully determines BER**
  - **In many cases, $Q \propto (\mu_1 - \mu_0)$**





Q>6: A rule of thumb!

•Correlation between OSNR and BER

•Given OSNR, an empirical formula to calculate BER for a single fiber is

$$\log_{10}(BER) = 10.7-1.45*(OSNR)$$

Example:

Assume that OSNR = 14.5 dB

Then $\log_{10}(BER) = 10.7-1.45*(14.5) = -10.30$

Therefore

$$BER = 10^{-10.30} \approx 10^{-10}$$

More complicated calculations may refer to CISCO webpage
http://www.ciscopress.com/articles/article.asp?p=30886&seqNum=5

$Q_{dB}=OSNR+10\log(BW_o/Bw_e)$,

$1/OSNR=1/OSNR_1+1/OSNR_2+\ldots\ldots$

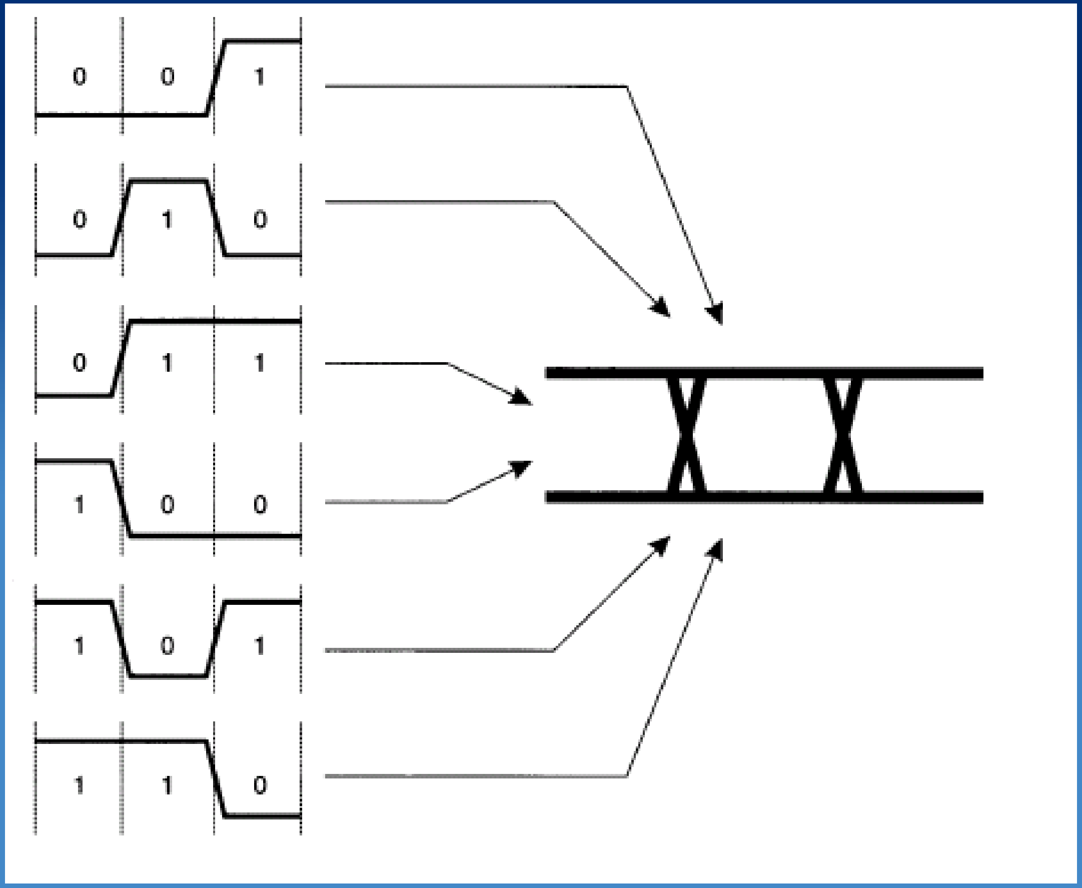
•Eye Diagram

•A measure of the quality and integrity of the electronic signal (since optical has been converted to electrical) is a ***superposition of bit periods on an oscilloscope***.

•If the signal has little noise and the amplitude is sufficient to be clearly recognized as “one” (marks) or “zero” (spaces) the superposition provides an “open eye”, otherwise eye is corrupted and “fuzzy”

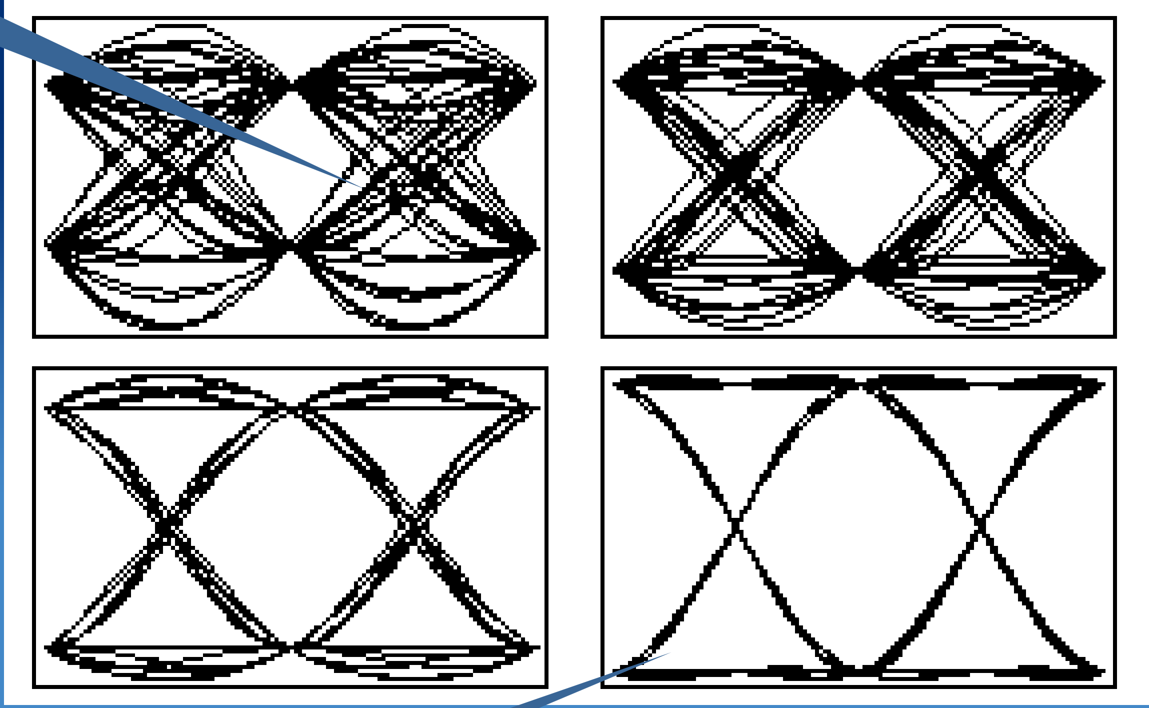
Eye Diagram (Part of Time domain Visualizer)

An **eye pattern** is obtained by superimposing the actual waveforms for large numbers of transmitted or received symbols



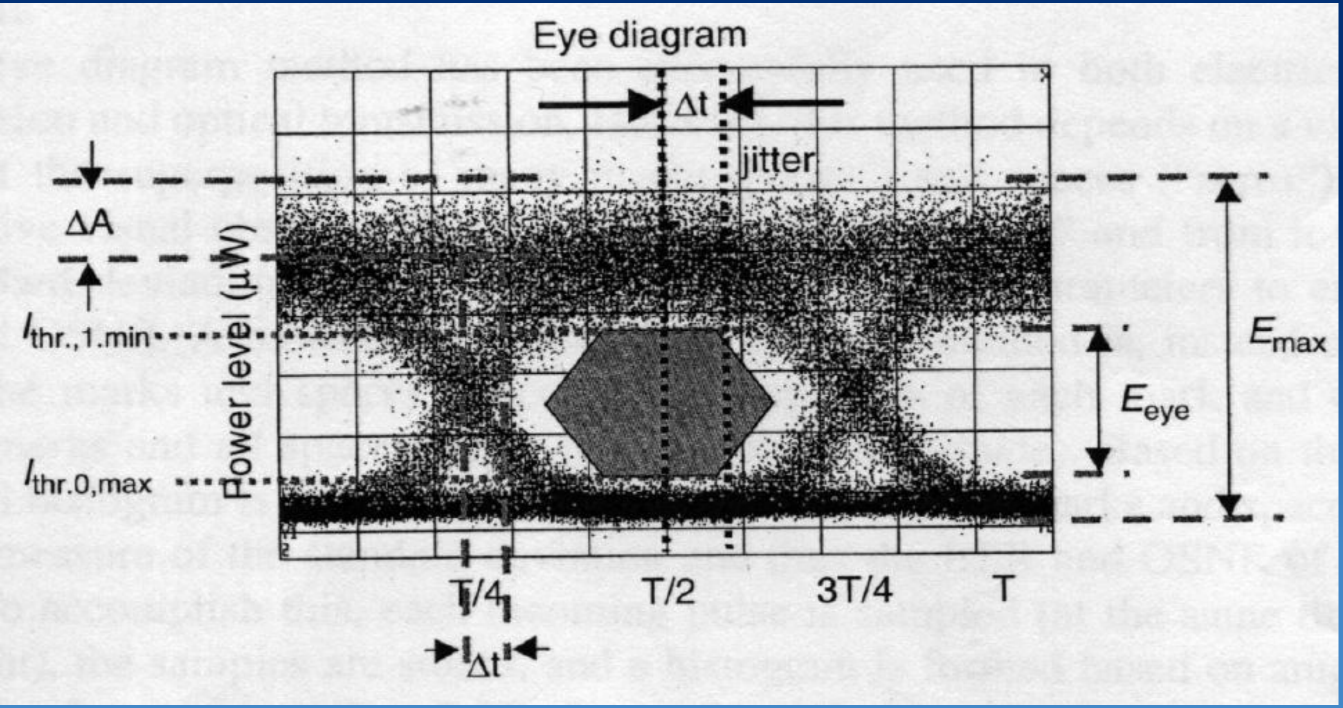
Eye Diagram (Part of Time domain Visualizer)

- **Noisy Eye Diagrams**



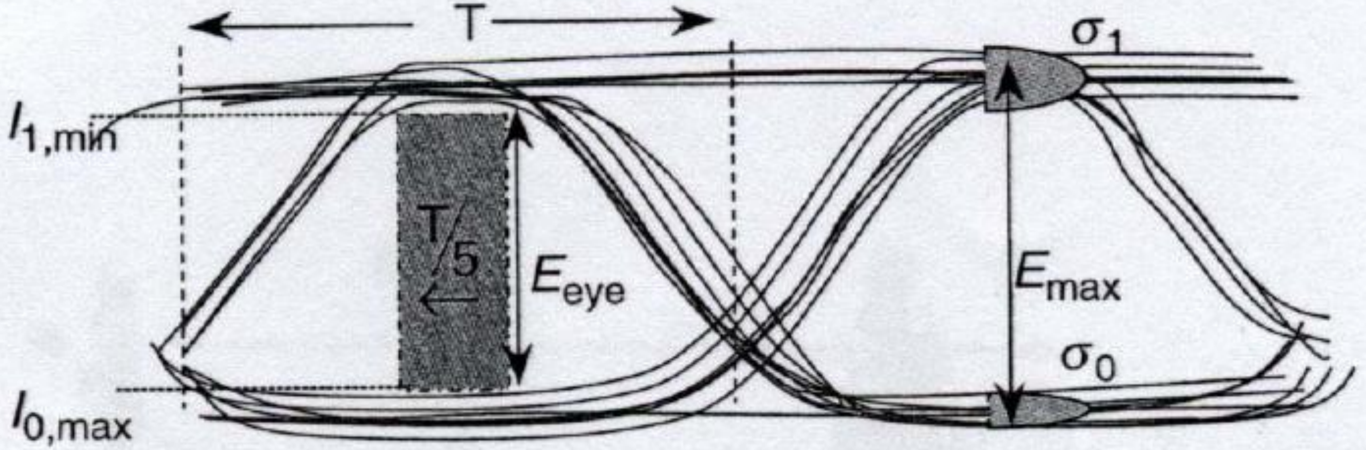
- **Low-noise Eye Diagrams**

•Eye Diagram



- Jitter: Δt
- Eeye， Emax

•Eye Diagram and Q-factor



1. Measure/estimate:

Variance for "0", Var_0

Variance for "1", Var_1

σ_0 = std deviation for "0" $= \sqrt{Var_0}$

σ_1 = std deviation for "1" $= \sqrt{Var_1}$

E_{eye}, E_{max}, E_{avg}

2. Calculate:

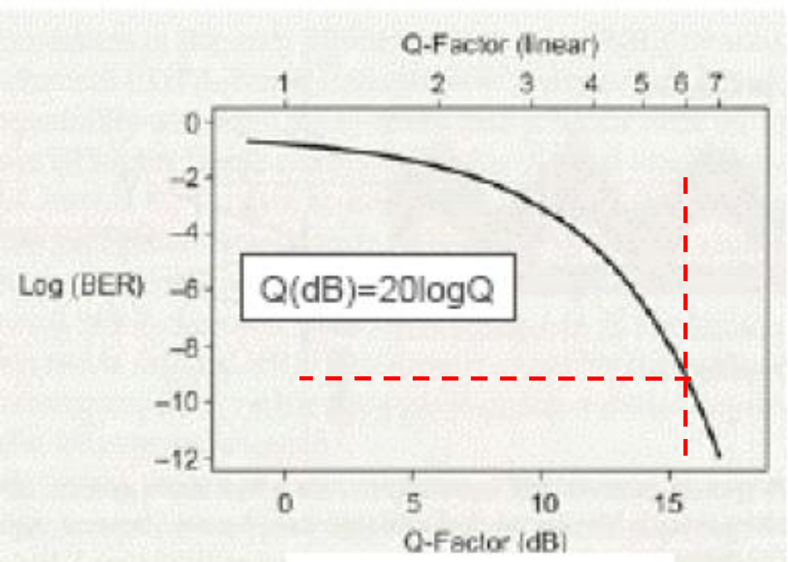
$E_{eye} = I_{1,min} - I_{0,max}$

$Q = E_{max} / \sqrt{\sigma_1^2 + \sigma_0^2}$

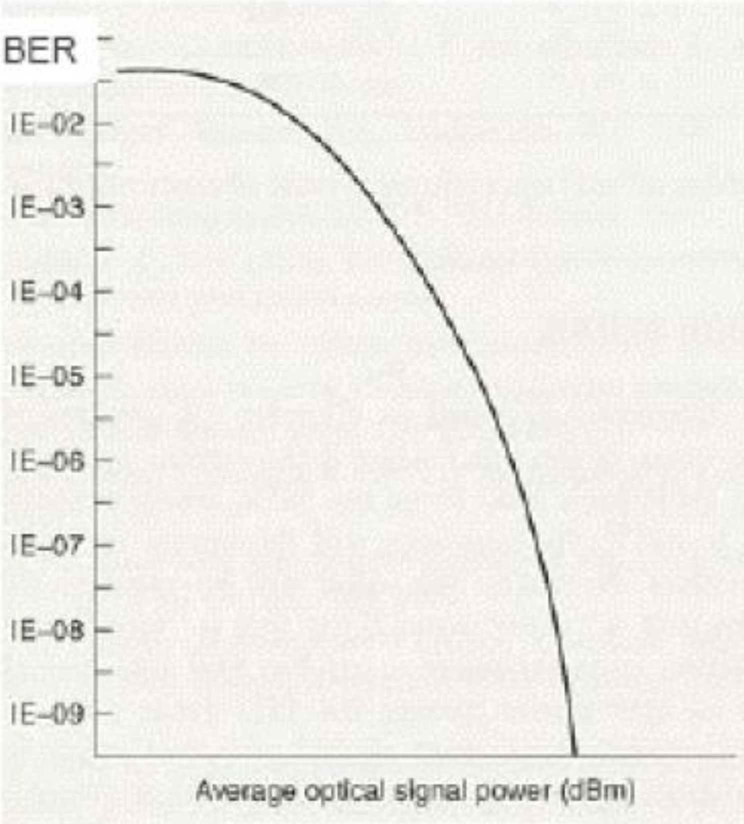
BER = ½ erfc $(Q/\sqrt{2})$

Penalty = $E_{eye}/2E_{avg}$

•BER and Q factor



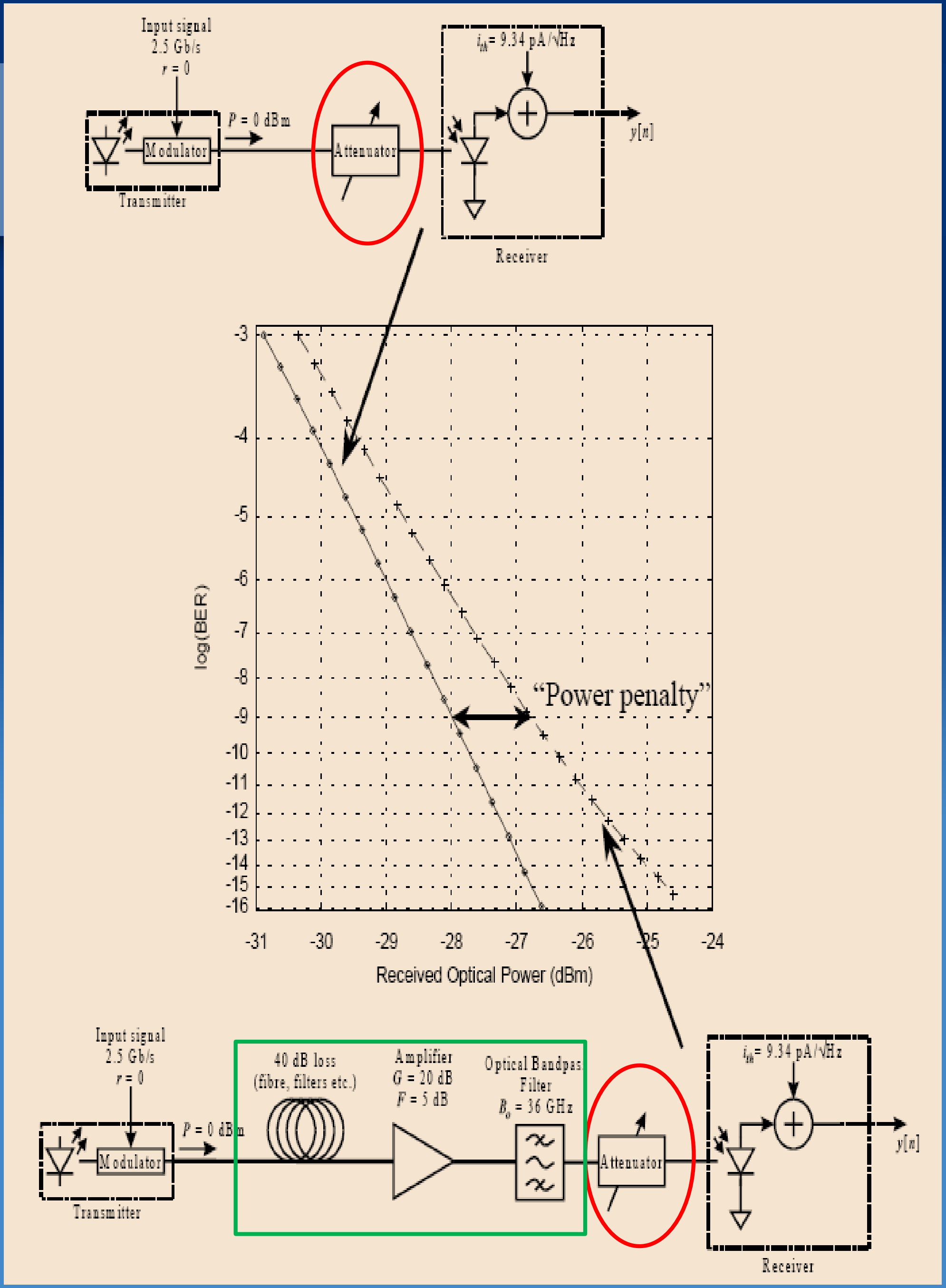
BER vs. Q-factor



BER vs. $P_{received}$

•Power penalty

- **Measure BER as a function of mean *received optical power (ROP)***
  - **For each ROP, decision threshold is optimized, and the BER is measured**
  - **Receiver sensitivity is the ROP required to achieve a specific BER (10^{-9} ,10^{-12})**
  - **•Find power penalty for with and without “optically amplified link”**



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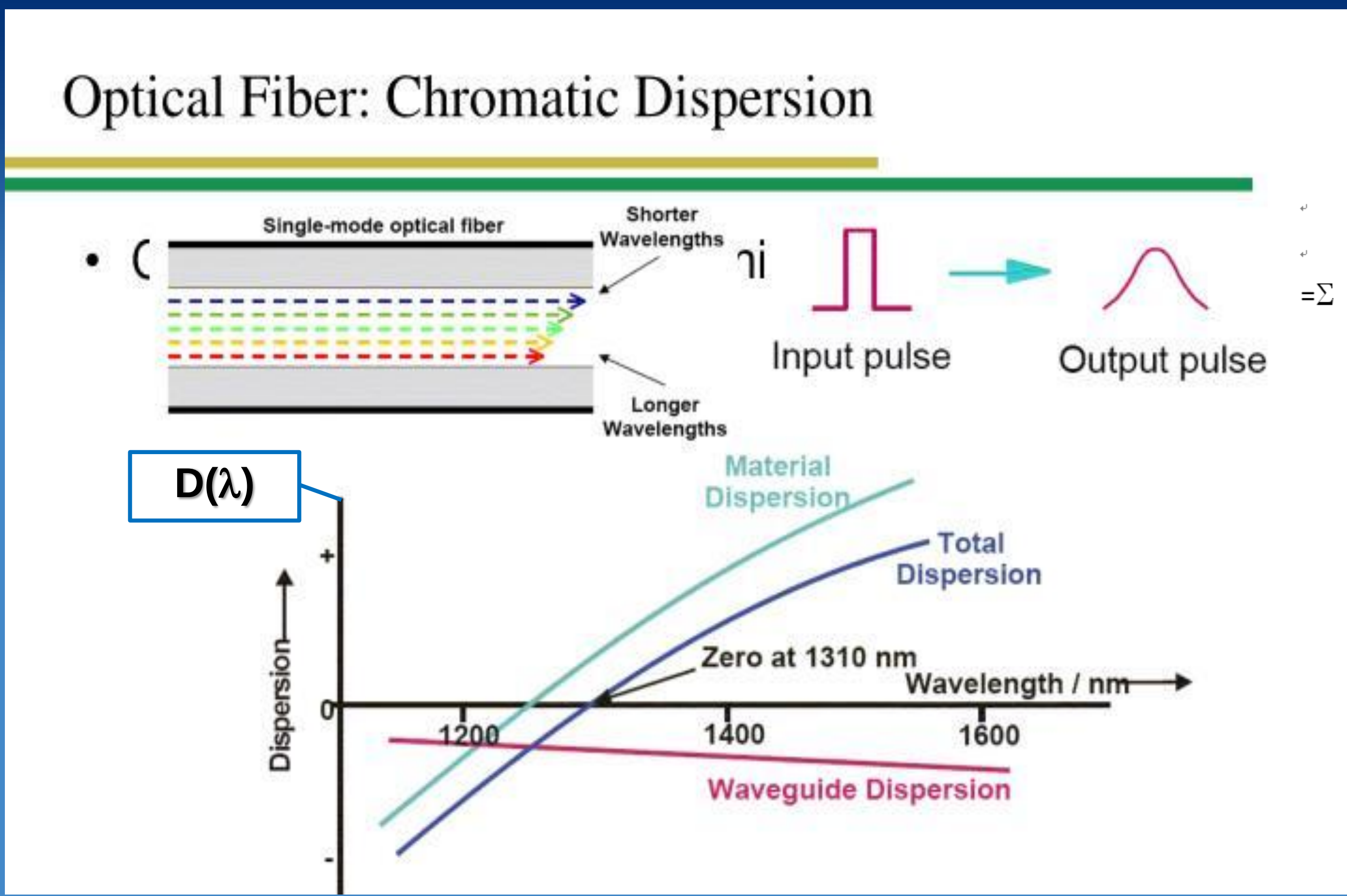
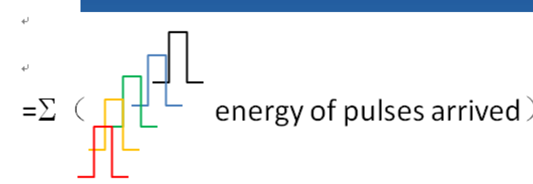
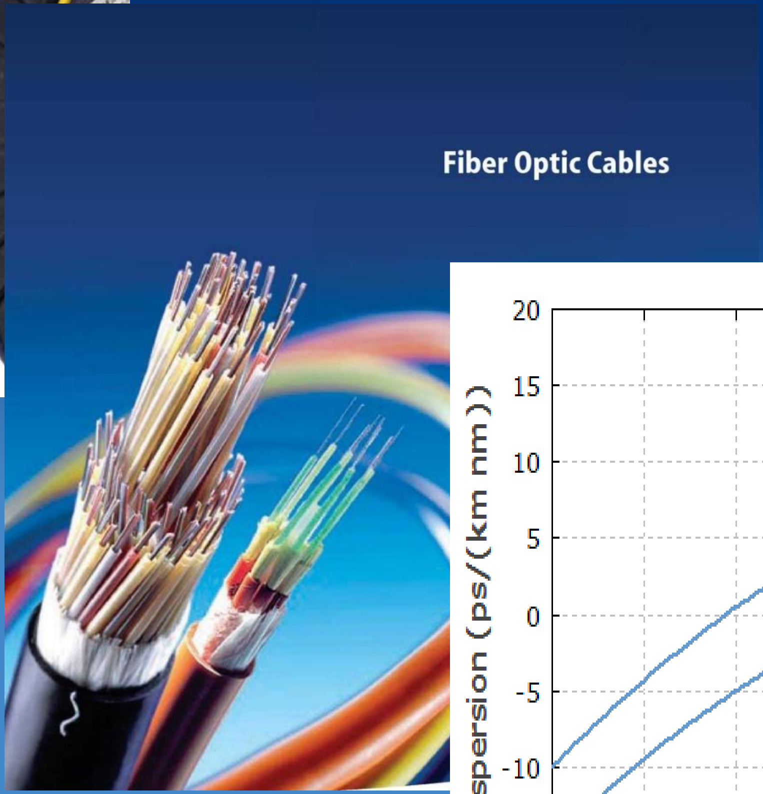
Dispersion in optical fiber

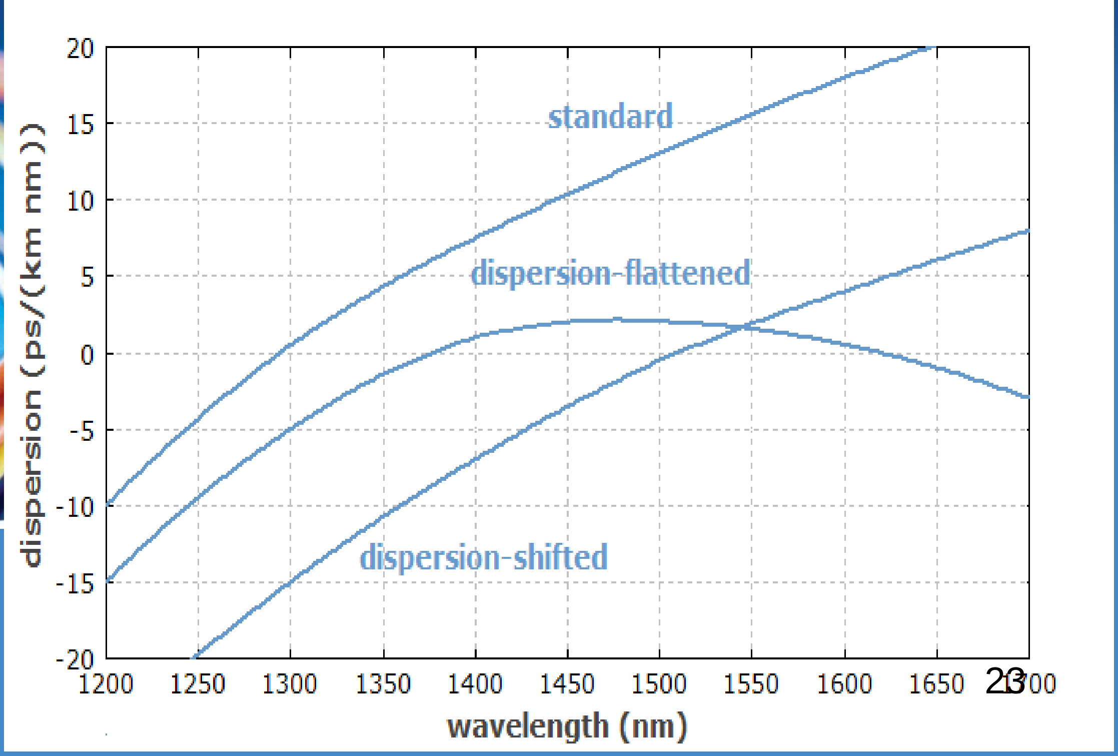
- All light sources have some degree of spectral width: $\Delta\lambda$.
- $n=n(\omega)$ or $n=n(\lambda)$ leading to different arrival time of a light pulse $\rightarrow$ pulse spread
- Pulse broadening: $\Delta T= D(\lambda)*\Delta\lambda*L$(distance)
- *D* is *dispersion parameter* and is expressed in units of ps/(km nm).
- Total dispersion = material dispersion + waveguide dispersion (for SMF)



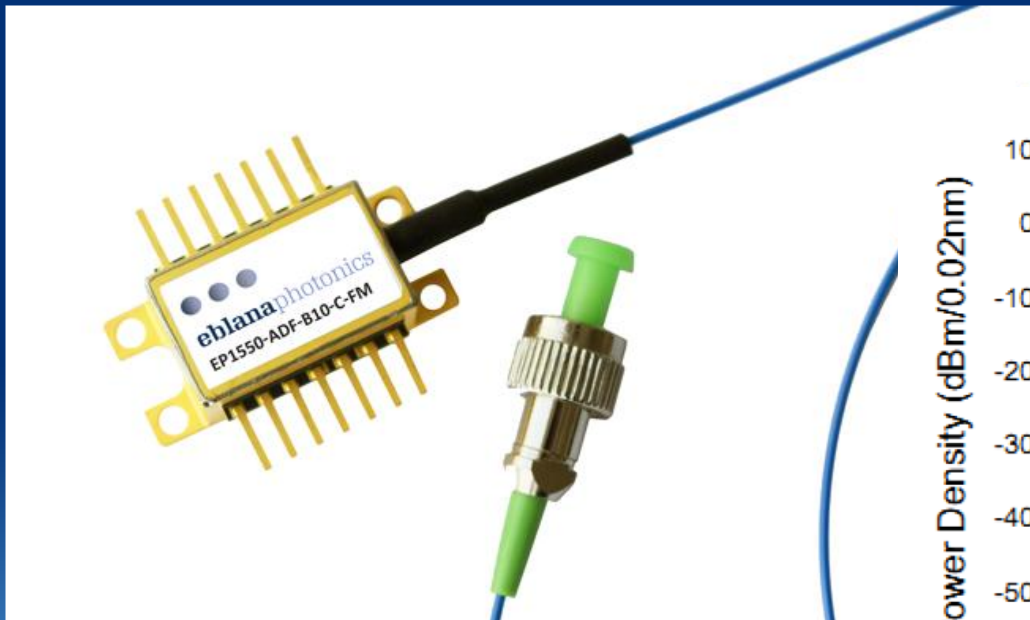


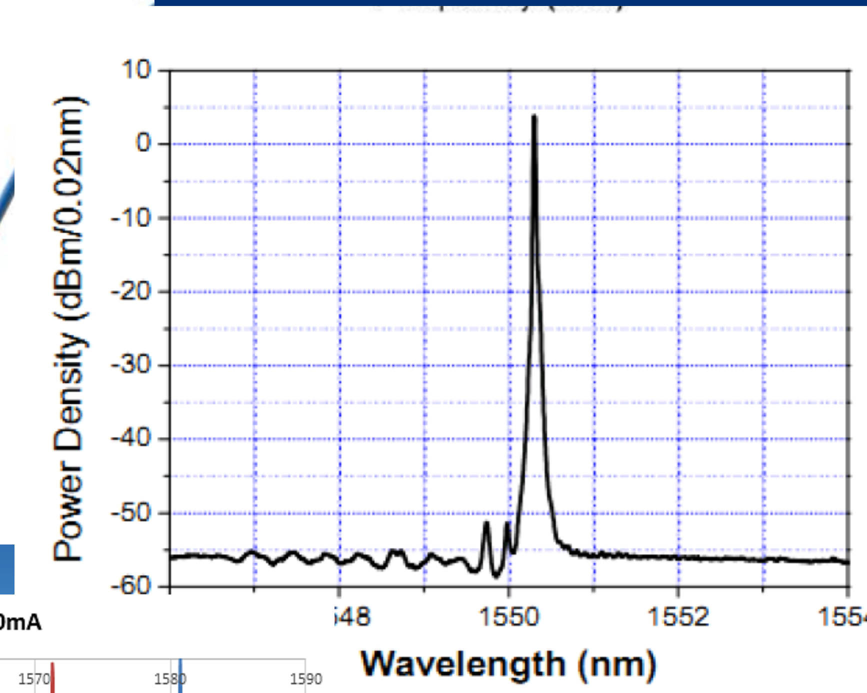
Dispersion of Commercial Optical Fibers ~10ps/km*nm

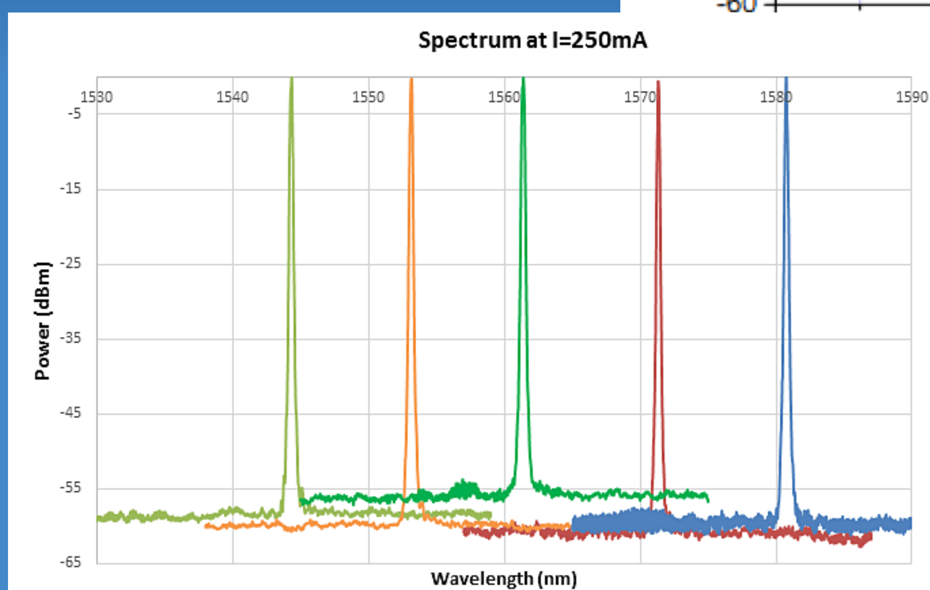




Laser Source for Optical Communications

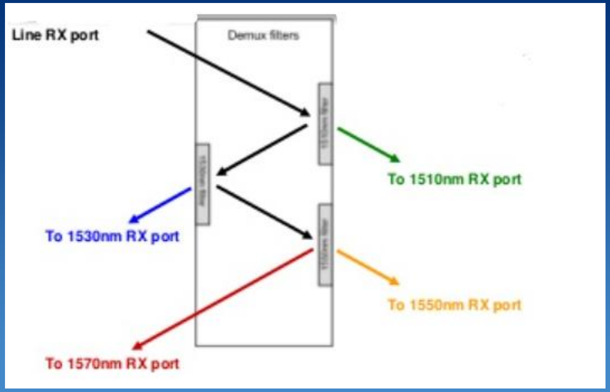




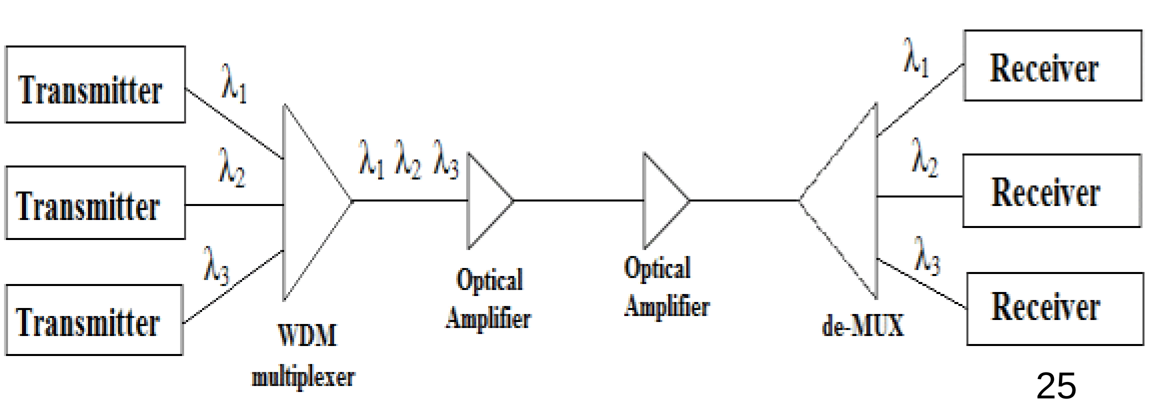


•Network Classification

- •Classified by Channel Spacing
  - •Wide : WWDM
  - •Coarse: CWDM 20nm
  - •Narrow: NWDM
  - •Dense: DWDM 100GHz, 0.8nm



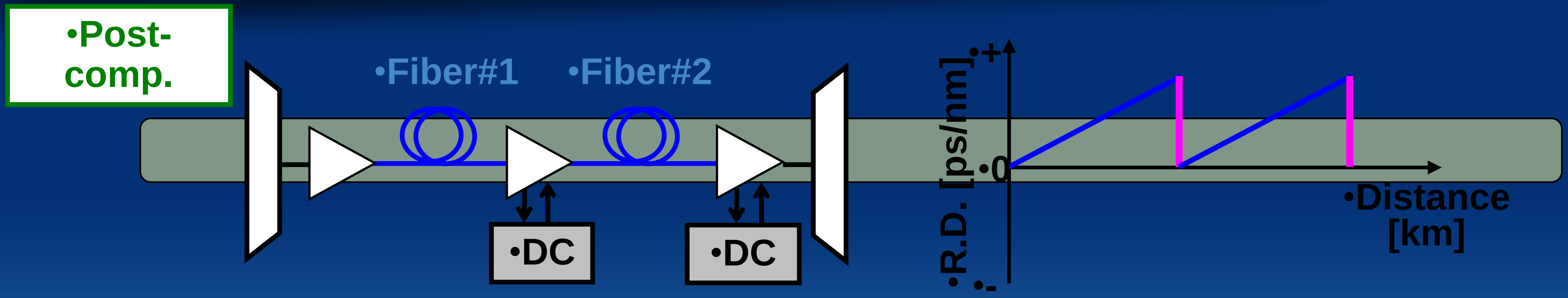
- •Classified by Interfaces
  - •Open
  - •Half-Open
  - •Integrated
- •Classified by
  - •Channel Numbers
  - •Channel Speed

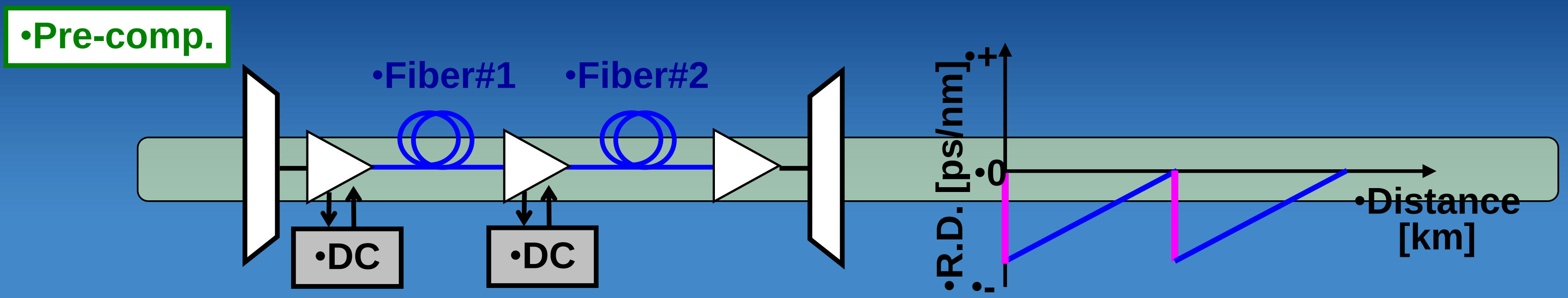


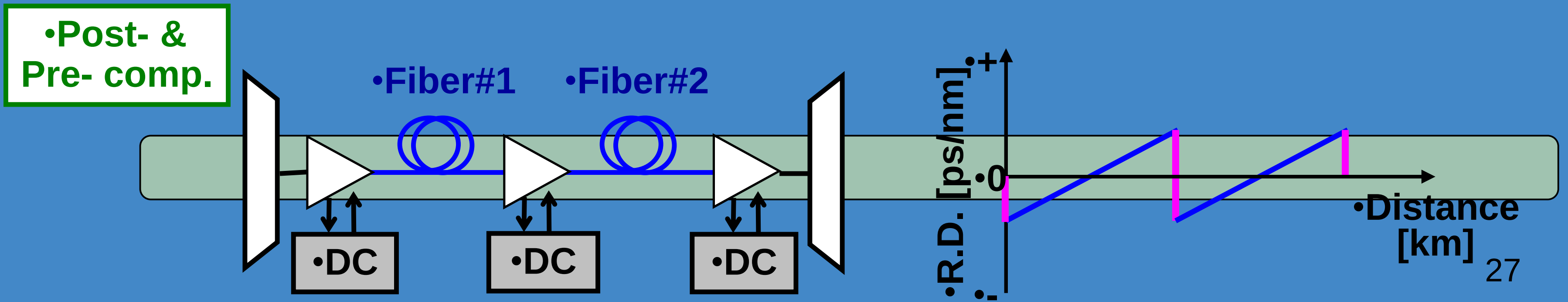
•Effects on WDM by A number of Factors

- Attenuation
  - Absorption and Scattering
- Dispersion
  - Modal, Waveguide, Material, …
- Signal-to-Noise ratio
  - ASE in EDFA
  - Accumulated Noise
- Nonlinear Effects
  - SPM, XPM, FWM, SBS, ….

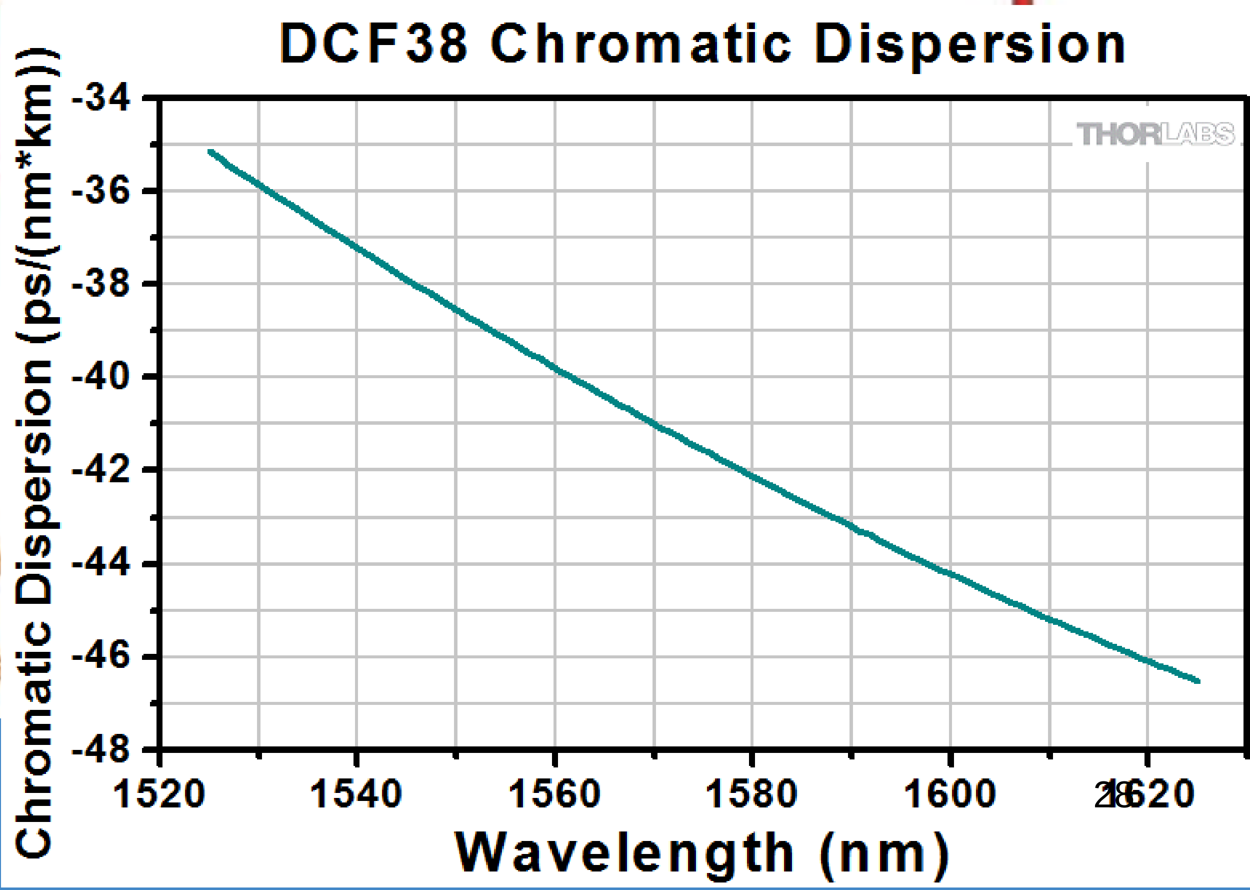
•Three compensation schemes





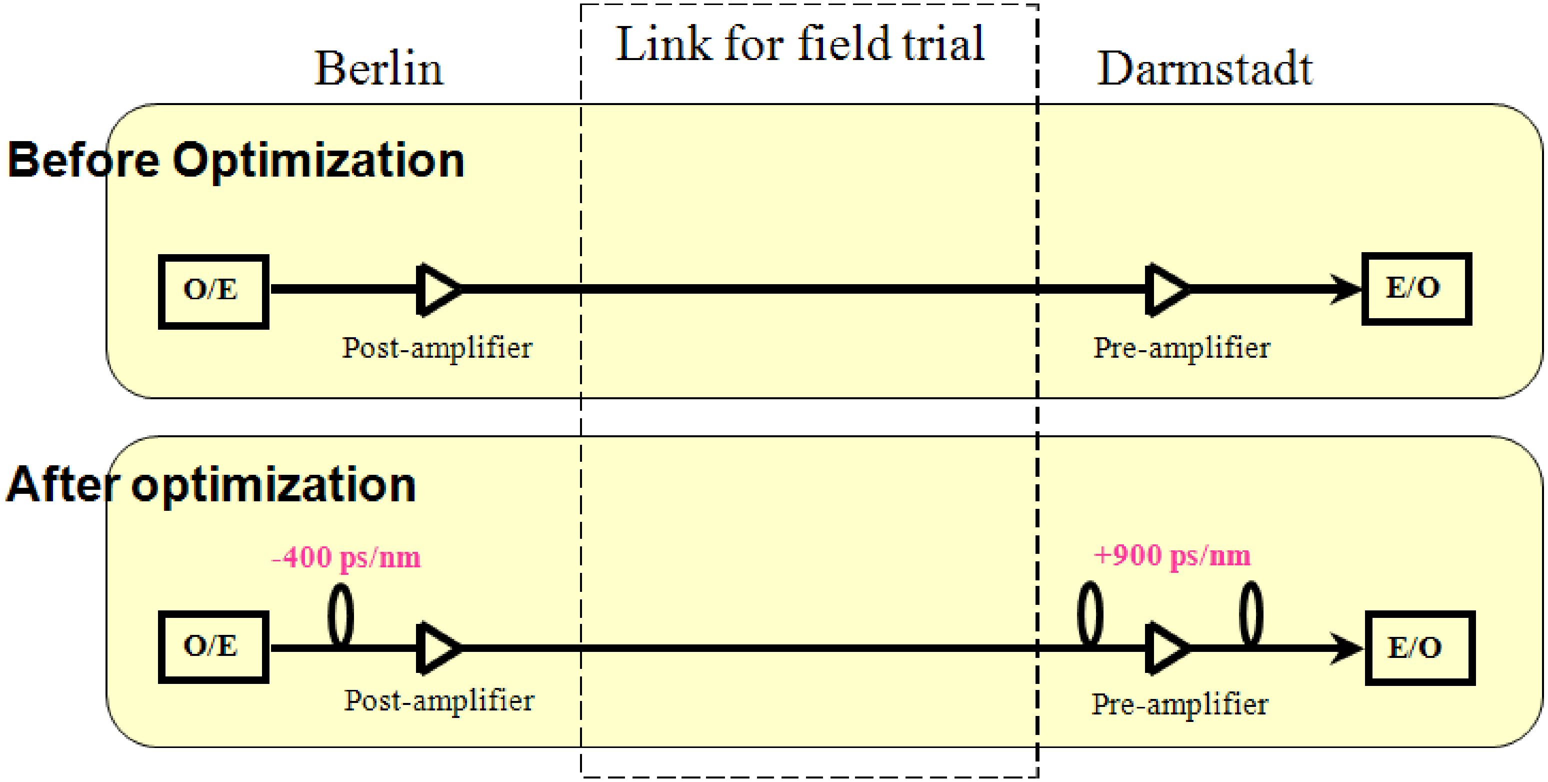


Dispersion Compensation Fiber (DCF)



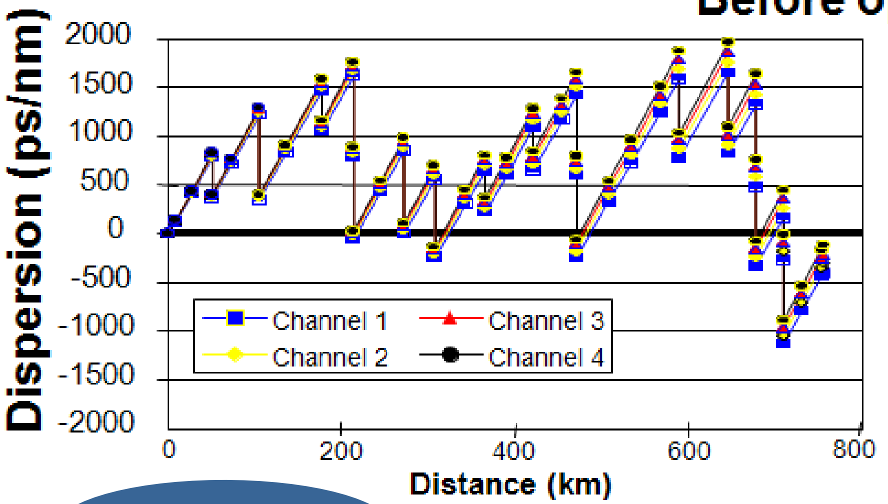
•A Field Example

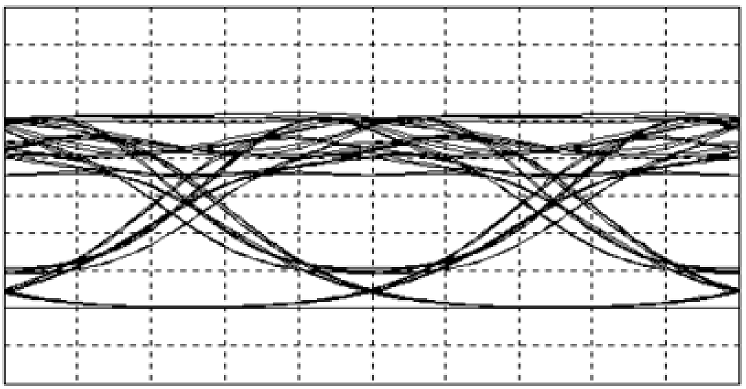
10Gbit/s 750km WDM field trial between Berlin and Darmstadt (Ref.: OFC/IOOC'99, Technical Digest TuQ2, A. Ehrhardt, et.al.)



•Before and After Optimization

Before optimization





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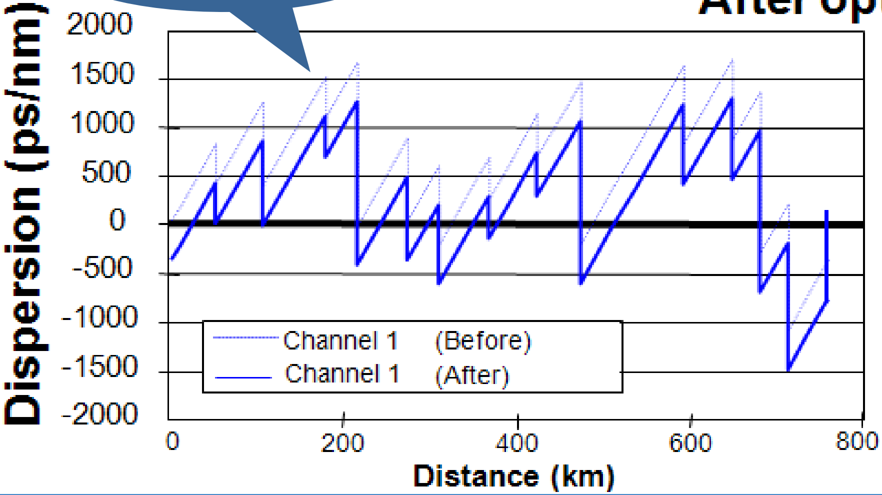
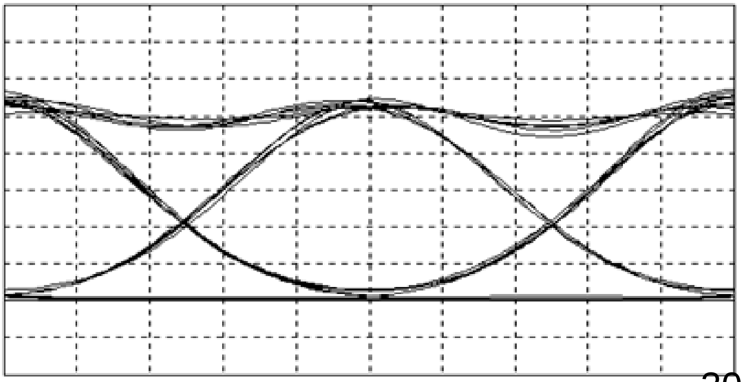
After optimization





•Fiber transmission - The Nonlinear Schrödinger Equation

•Light in an optical fiber

$$\vec{E}(\vec{r},t) = F(x,y)\,\boxed{E(z,t)}\,e^{i(\beta_0 z-\omega_0 t)}\,\hat{u}$$

•Transverse field profile defined by mode

•slowly varying component

•carrier

•polarization

•The nonlinear Schrödinger equation (NLSE)

$$\frac{\partial E}{\partial z} = -\frac{\alpha}{2}E - i\frac{\beta''}{2}\frac{\partial^2 E}{\partial t^2} + i\gamma|E|^2 E$$

•Launched signal: $E(0,t)$

•C. R. Menyuk, IEEE J. Quantum Electron. 25, 12, 2674-2682, December 1, 1989.

•L. F. Mollenauer *et al.*, in *Opt. Fiber Telecommunications iVA*, (Academic, San Diego, Calif., 1997).

•Fiber transmission –NLSE general definitions

$$\frac{\partial E}{\partial z} = -\frac{\alpha}{2}E - i\frac{\beta''}{2}\frac{\partial^2 E}{\partial t^2} + i\gamma|E|^2 E$$

•Attenuation •Dispersion •Nonlinearity

$$n = n_0(\omega) + n_2 I$$

α - Loss coefficient [km^{-1}]

β'' - dispersion coefficient [ps^2/km]

$\gamma = 2\pi n_2/(\lambda A_{eff})$ - nonlinear coefficient [W^{-1}km^{-1}]

$|E|^2$ **•- is the optical power [W]**

•to remove propagation delay

•$t = t_{old} - z/v_g$

•Fiber transmission – Renormalizing the losses

•Define:

$$a(z,t) = E(z,t)\exp\left(\frac{\alpha}{2} z\right)$$

•Loss normalized NLSE

$$\boxed{\frac{\partial a}{\partial z} = -i\frac{\beta''}{2}\frac{\partial^2 a}{\partial t^2} + i\gamma G(z)|a|^2 a}$$

$$\beta'' = \frac{d^2\beta}{d\omega^2} = \frac{d}{d\omega}\left(\frac{1}{v_g}\right)$$

Unit: ps^2/km

$$\boxed{D = -\frac{2\pi c}{\lambda^2}\beta''}$$

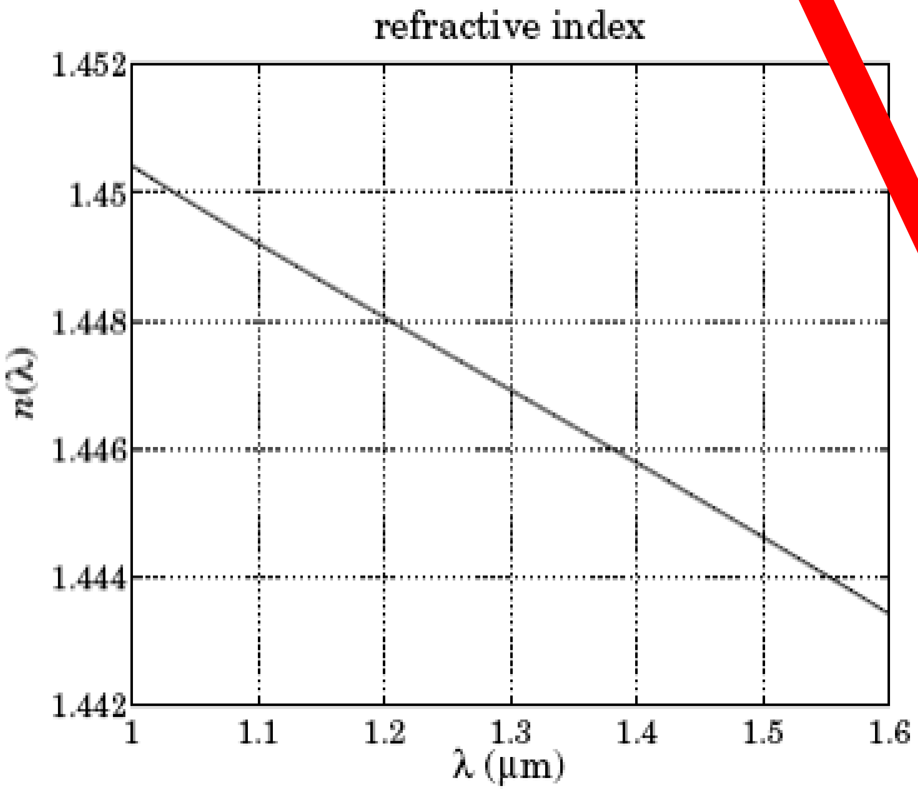
Unit: ps/(km nm)

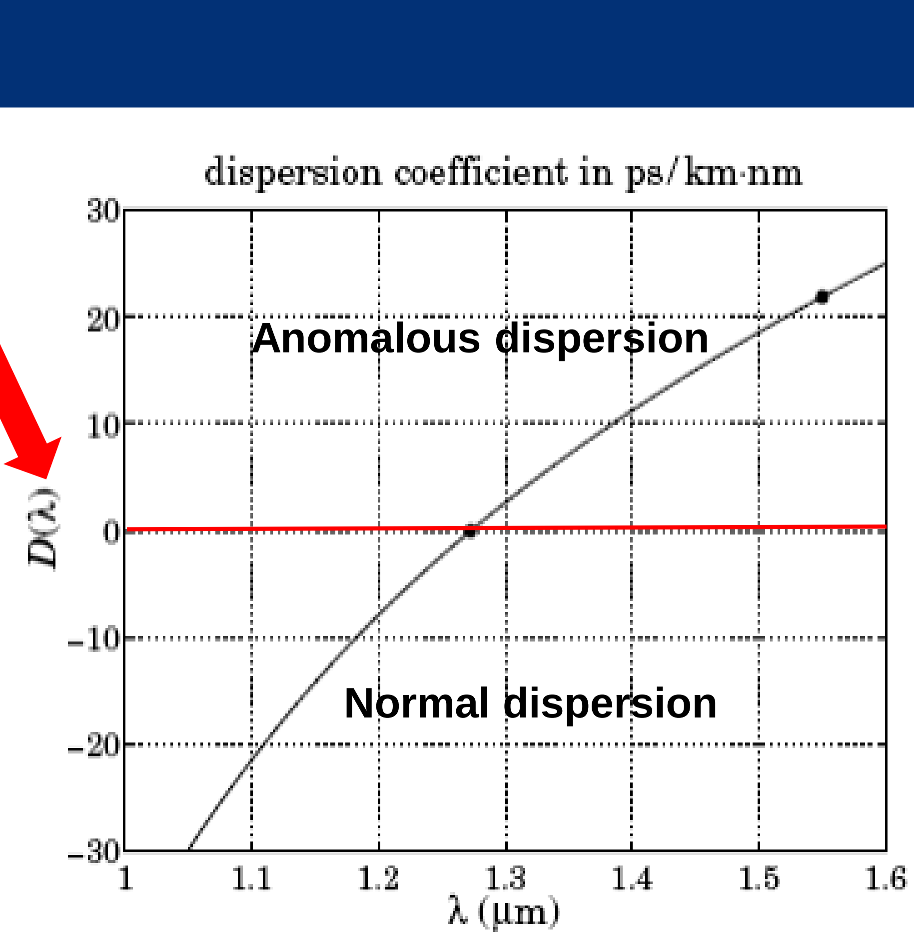
•"Positive" or "negative" dispersion depends on the definition

•D > 0 $\Rightarrow \beta'' < 0$ "anomalous" dispersion: $\lambda\uparrow$, $v_g\downarrow$

•D < 0 $\Rightarrow \beta'' > 0$ "normal" dispersion: $\lambda\uparrow$, $v_g\uparrow$

•Dispersion coefficient D



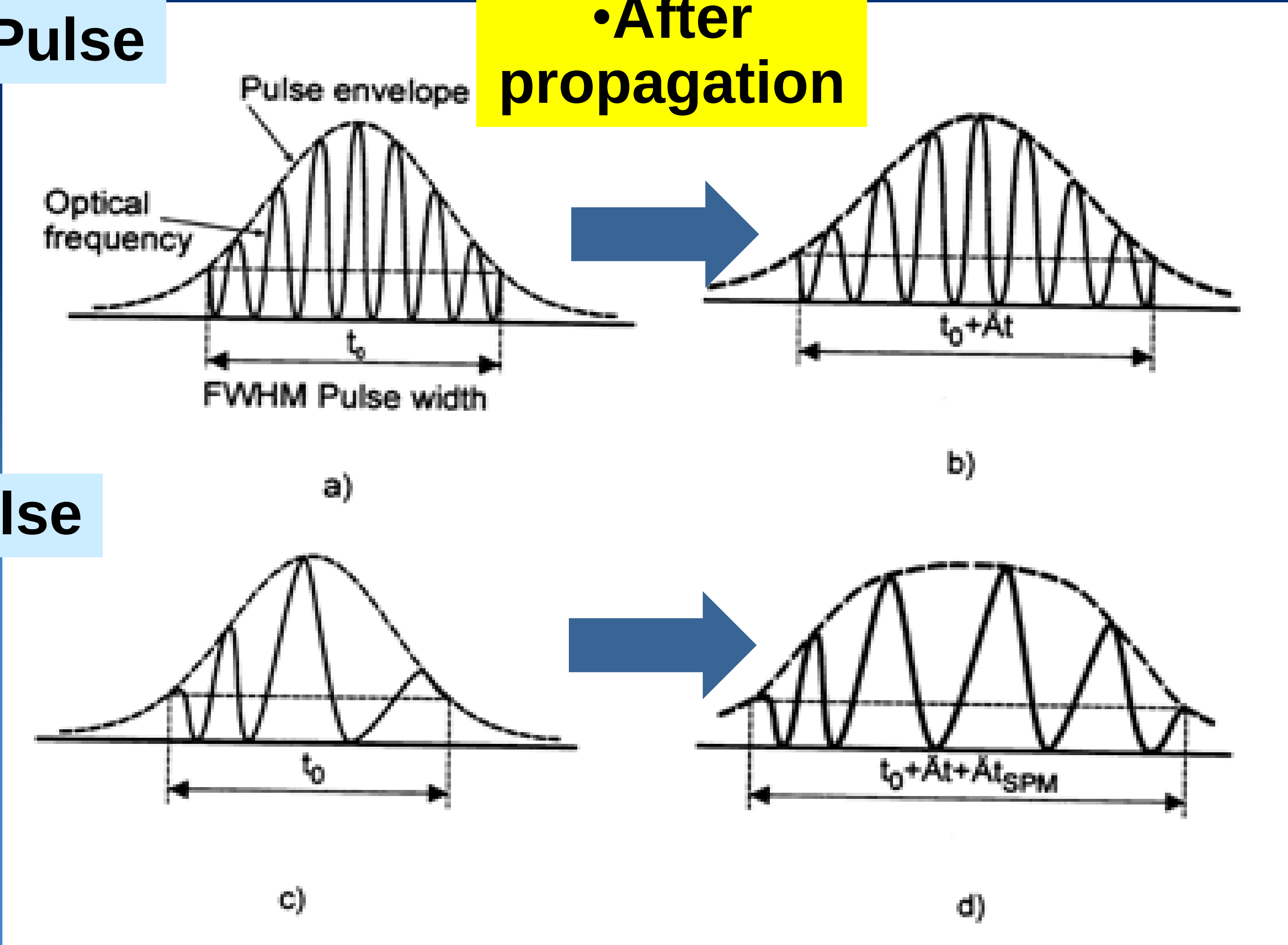


•Group velocity dispersion (GVD)

•Unchirped Pulse

•After propagation

•Chirped Pulse



•Linear propagation of pulses

•Negligible nonlinearity low power ($|a|^2$ is small)

$$\frac{\partial a}{\partial z} = -i\frac{\beta''}{2}\frac{\partial^2 a}{\partial t^2} + \cancel{i\gamma G(z)|a|^2 a}$$

•Trivial solution in the Fourier domain

$$\frac{\partial \tilde{a}}{\partial z} = i\frac{\beta''}{2}\Omega^2\tilde{a}$$

$$\Omega = \omega - \omega_0$$

$$\tilde{a}(\Omega, z) = \boxed{\exp(i\beta''\Omega^2 z/2)}\,\tilde{a}(\Omega,0)$$

•Three things happened after propagation for a Gaussian pulse

•Time domain

$$a(t,0) = A_0 \exp\left[-\frac{t^2}{2\tau_0^2}\right]$$

•FT

•Frequency domain

$$\tilde{a}(\Omega,0) = A_0\sqrt{2\pi\tau_0^2}\, e^{-\frac{\Omega^2}{2}\tau_0^2}$$

•exp($i\beta''\Omega^2 z/2$)

$$\tilde{a}(\Omega,z) = A_0\sqrt{2\pi\tau_0^2}\, e^{-\frac{\Omega^2}{2}(\tau_0^2 - i\beta'' z)}$$

•IFT

$$a(t,z) = A(z) e^{-\frac{1+iC(z)}{2\tau(z)^2}t^2}$$

Where: $C(z) = \mathrm{sign}(\beta'')z/L_D = \beta'' z/\tau_0^2$ **is the chirp parameter**

Complex pulse-width is associated with chirp

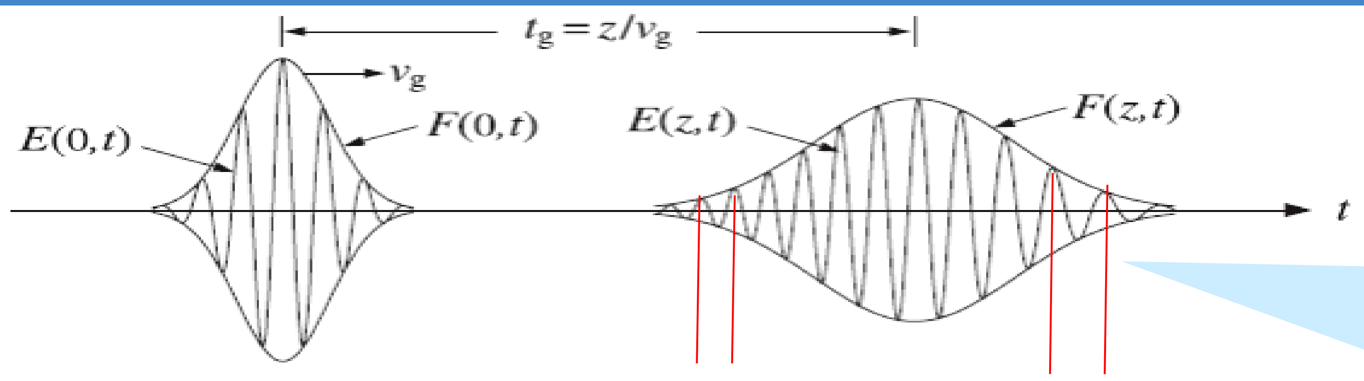


Fig. 3.6.1 Pulse spreading and chirping.

1. **Amplitude down;**
2. **Pulses spread = D×Δλ×L**
3. **Chirp**

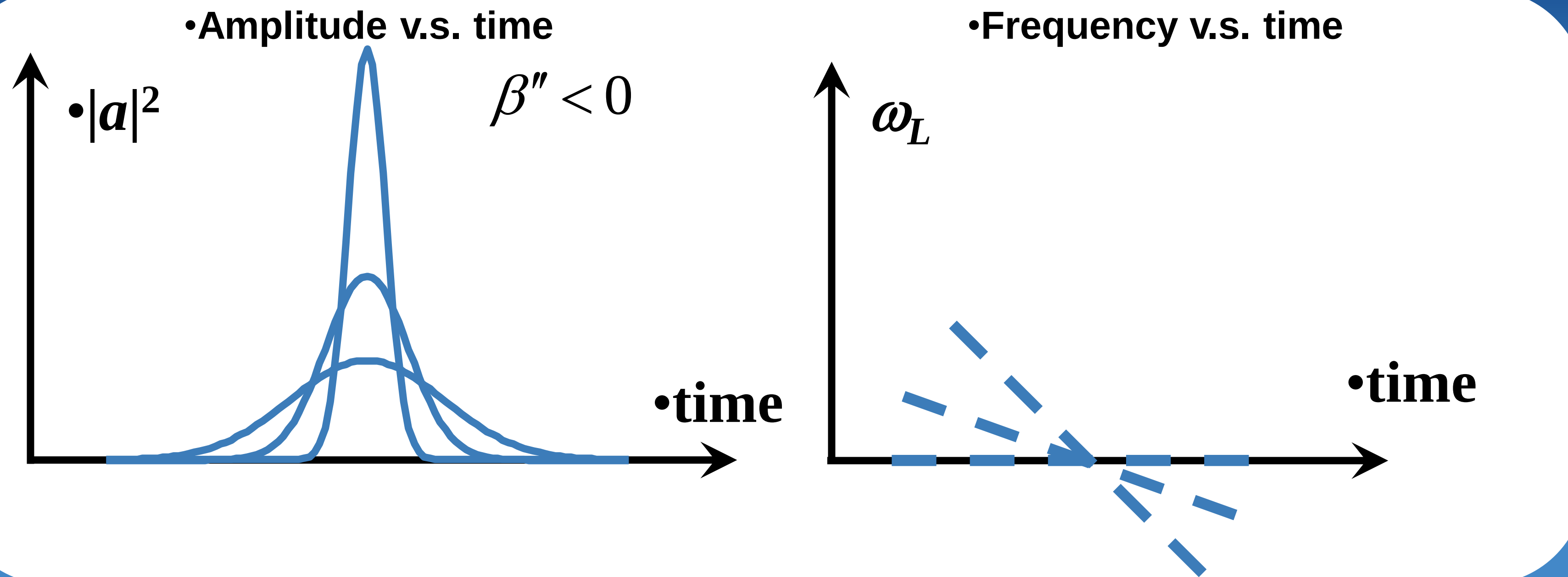
•The chirp of a pulse

$$a(t,z) = A(z)\exp\left[-\frac{t^2}{2\tau(z)^2} + i\varphi_L(z,t)\right]$$

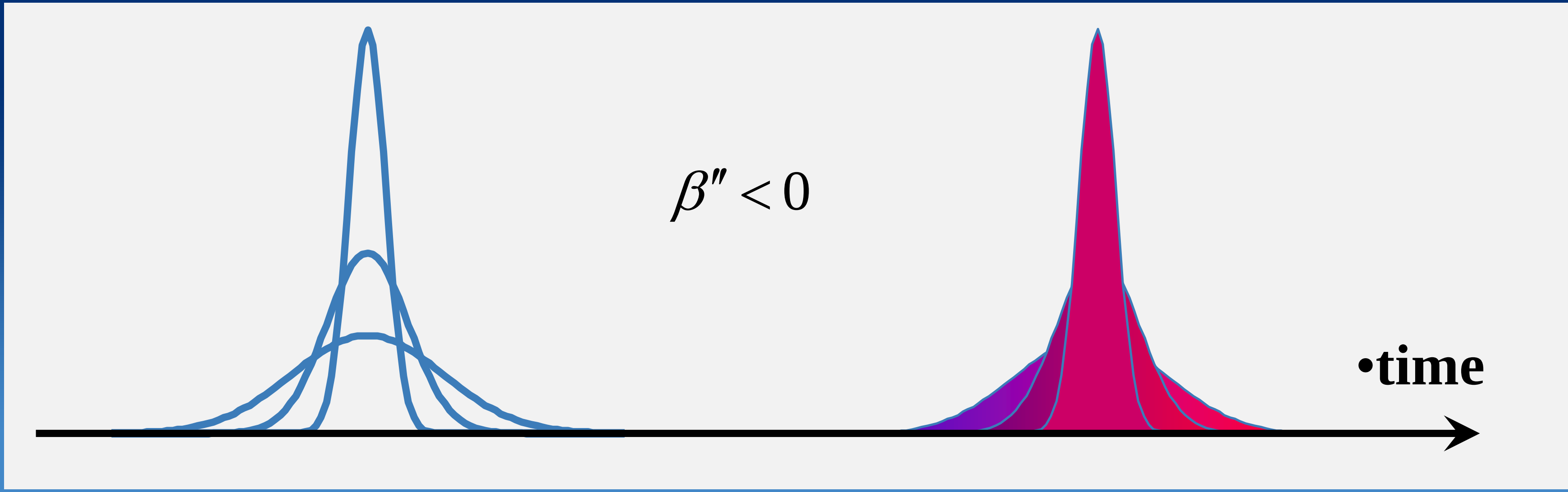
$$\varphi_L = \frac{C(z)}{2\tau(z)^2}t^2$$

•instantaneous frequency

$$\omega_L(t,z) = -\frac{\partial\varphi_L}{\partial t} = \operatorname{sign}(\beta'')\frac{z/L_D}{\tau(z)^2}t$$



•The chirp of a pulse: shifting colors



•If $\beta'' < 0$, the blue part of the pulse goes faster and arrives earlier

•Propagation of pulses: Zero-dispersion case

$$\frac{\partial a}{\partial z} = \cancel{-i\frac{\beta''}{2}\frac{\partial^2 a}{\partial t^2}} + i\gamma|a|^2 a$$

$$\Downarrow$$

$$\frac{\partial a}{\partial z} = i\,\gamma|a|^2 a$$

•Time is a parameter. There is a closed-form solution.

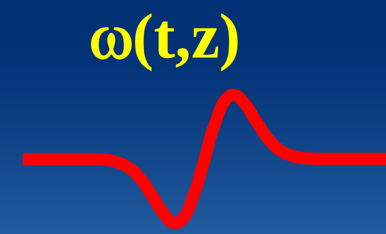
$$\frac{\partial\left(|a(t,z)|^2\right)}{\partial z} = 0 \Rightarrow |a(t,z)|^2 = |a(t,0)|^2 \Rightarrow a(t,z) = e^{i\gamma|a(t,0)|^2 z}a(t,0)$$

•Self phase modulation

•Nonlinear → The pulse acquires a chirp!!!

$$a(t,z)=\exp\left[i\gamma\left|a(t,0)\right|^{2}z\right]a(t,0)=\exp\left[i\varphi_{\mathrm{NL}}(t,z)\right]a(t,0)$$

$$\omega_{\mathrm{NL}}(t,z)=-\frac{\partial\varphi_{\mathrm{NL}}}{\partial t}=-\gamma\frac{d\left|a(t,0)\right|^{2}}{dt}z$$

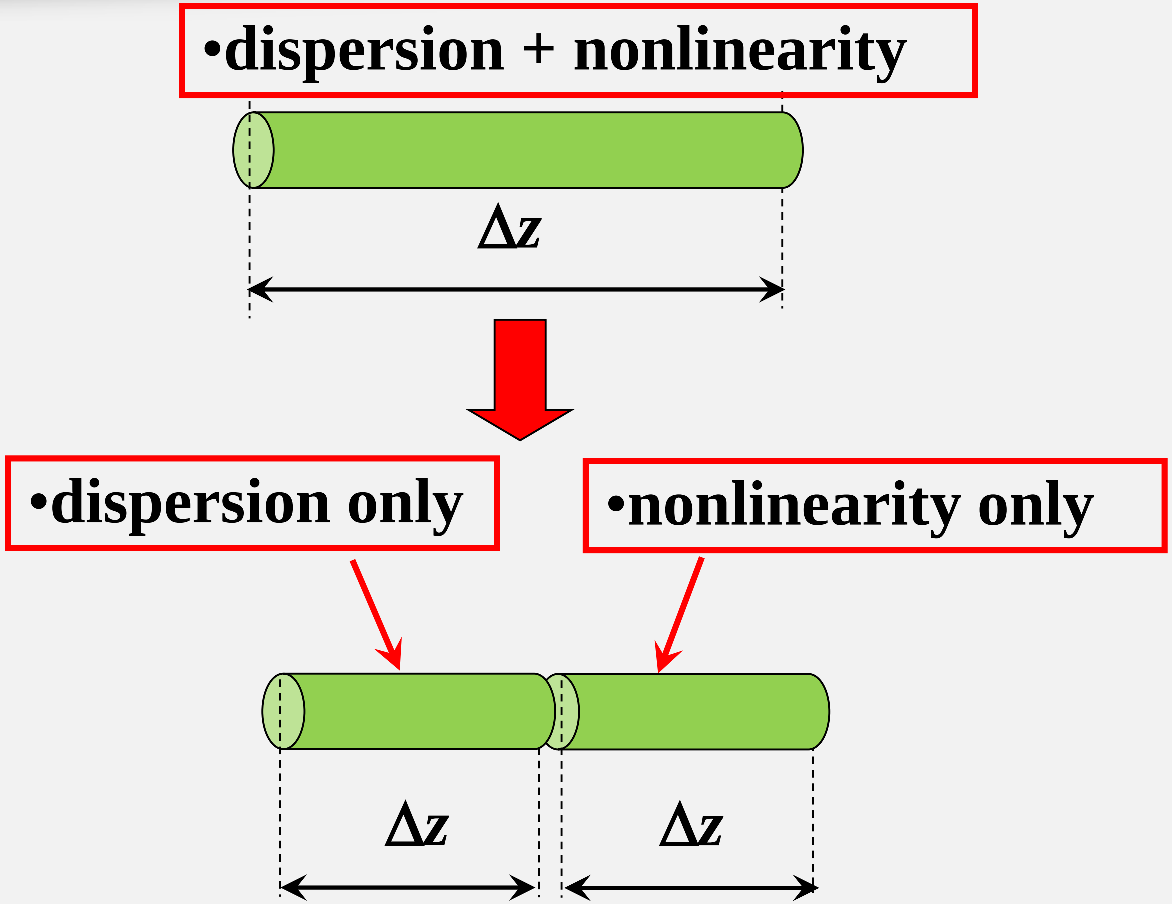


•The pulse leading edge shifts to the red and the trailing to the blue

•The pulse acquires a chirp, <u>amplitude profile does not change</u>.

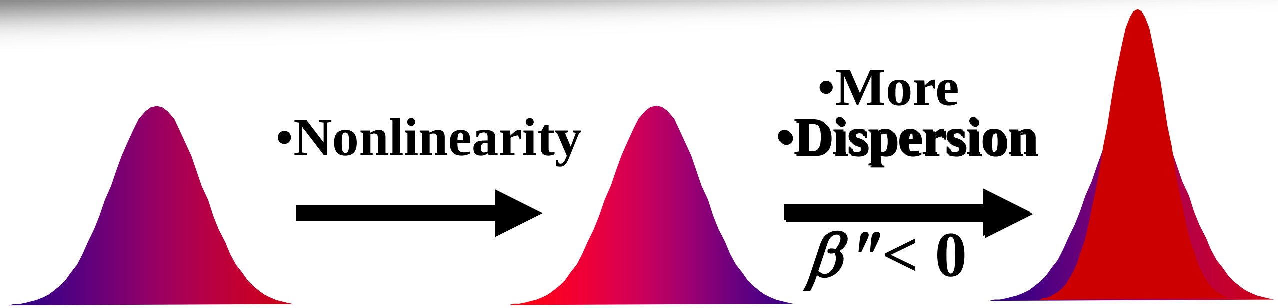
•Required bandwidth increases: b/c generation of new frequencies

•Dispersion + nonlinearity: The split-step method



•G. P. Agrawal, *Nonlinear Fiber Optics*, Third Edition, (Academic, New York, 2001).

•Balancing dispersion and nonlinearity – a simplified example



•Nonlinearity: leading edge shifted to the red, trailing to the blue

$\beta'' < 0$: the blue part of the pulse travels faster than the red part leading to pulse compression and inversion of chirp

•Possible to achieve average <u>compensation</u> between nonlinear compression and the dispersive broadening $\Rightarrow$ Stable pulse propagation

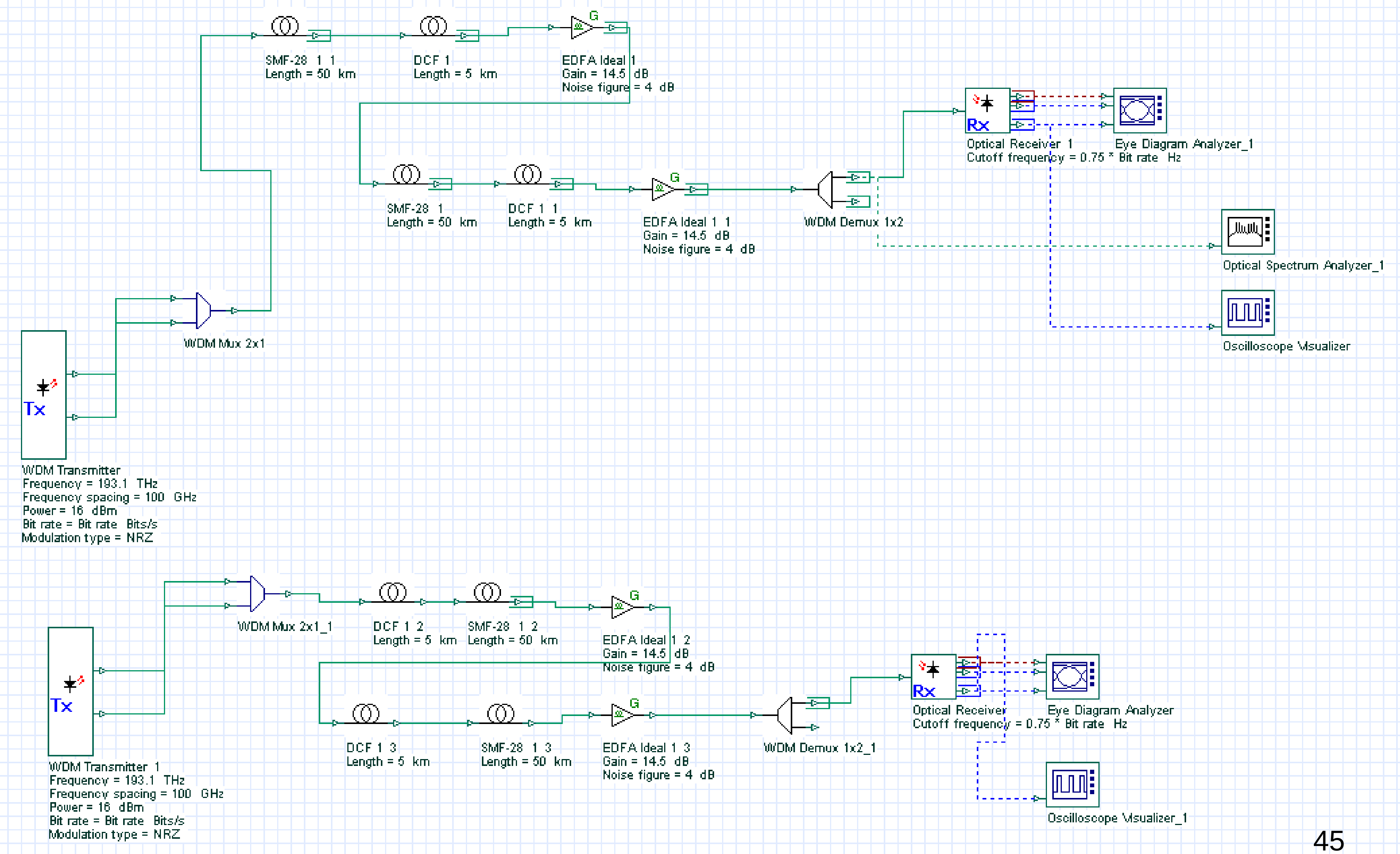
•Nonlinear dynamics of a single pulse

•If linear and nonlinear chirp compensate exactly after one dispersion map period ⇒ ~constant bandwidth.

•If linear and nonlinear chirp ***do not*** compensate exactly ⇒ bandwidth oscillations over a scale larger than the amplifier spacing

•Large gap between linear and nonlinear chirp ⇒ Pulse destroys ⇒ Dispersing pulses

•Today's Exercise: Pre- v.s. post-compensation



•Pre-compensation v.s. Post-compensation